

Unit V	Basics of Computer Network: Computer settings (like computer name, work group, IP, DNS, proxy server, password, users, etc.), Trouble shooting of the computer system. Need to share resources, Distributed System and types of Network (LAN, MAN, WAN), Network topologies (Bus, Star, Ring, Tree, Mesh) Components used to form network. Concepts of Layering	20%
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1. Computer Name

Definition:

A **computer name** is a unique identifier assigned to a computer within a network. It allows other devices and users to recognize and connect to it.

Importance:

Used to identify a device in local networks or domains.
Helps in file sharing, printer access, or remote access.
Makes remote troubleshooting and maintenance easier.

How to Find or Change:

To view:

Right-click *This PC* → *Properties* → *Computer Name, Domain, and Workgroup Settings* section.

Or

Open *Command Prompt* → type `hostname` → press *Enter*.

To change:

Control Panel → *System and Security* → *System* → *Change Settings* → *Computer Name tab*.

2. Workgroup

Definition:

A **workgroup** is a type of **peer-to-peer (P2P)** network model used mainly in **home or small office networks** where computers share files and devices directly without a centralized server.

Features:

All computers are peers (no hierarchy).

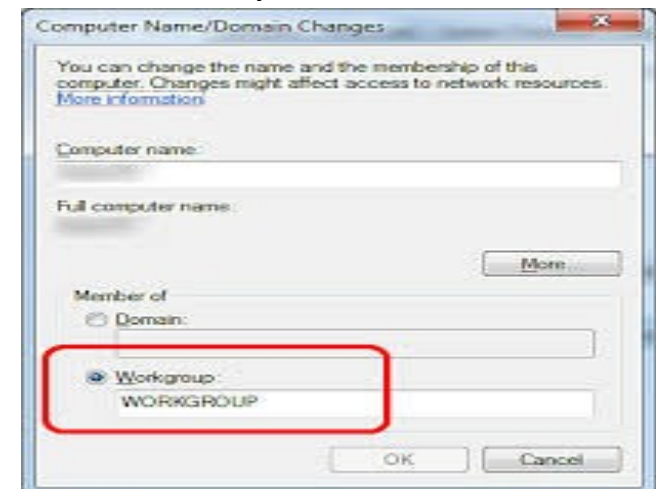
Each computer has its own user accounts.

Used for small-scale sharing of files, printers, and internet.

Usually part of the same **local network**.

Example:

If five computers are part of the same workgroup named *OFFICE-PC*, they can see each other under the *Network* section and share files easily.



Difference between Workgroup and Domain

Feature	Workgroup	Domain
Network Type	Peer-to-peer	Client-Server
User Accounts	Stored locally	Managed centrally on a server
Scale	Small	Large
Administration	Decentralized	Centralized
Best For	Home or small office	Large organizations

3. IP Address

Definition:

An **IP Address (Internet Protocol Address)** is a numerical label assigned to every device connected to a network that uses the Internet Protocol for communication.

Types:

IPv4: 32-bit address (e.g., 192.168.1.10)

IPv6: 128-bit address (e.g., 2401:4900:1f2a::1)

Role:

Uniquely identifies each device.

Ensures correct delivery of data packets from one device to another.

Commands:

ipconfig /all → displays complete IP configuration (Windows)

ping <IP> → checks connectivity with another device

Example:

If your PC has an IP of 192.168.0.5, and your printer has 192.168.0.10, both can communicate within the same network.

```
C:\WINDOWS\system32\cmd.exe
C:\Users\Lee>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

    Connection-specific DNS Suffix  . : mshome.net
    Link-local IPv6 Address . . . . . : fe80::d591:5dc9:72eb:2f3b%19
    IPv4 Address. . . . . : 192.168.137.249
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.137.1
```

4. DNS (Domain Name System)

Definition:

The **DNS** is the "phonebook of the Internet."

It converts **domain names** (like `www.google.com`) into **IP addresses** (like `142.250.182.110`) that computers use to locate each other.

Importance:

Allows users to use easy-to-remember names instead of numeric IPs.

Speeds up web browsing by caching domain name resolutions.

Managed by **DNS servers** provided by ISPs or configured manually.

How It Works:

User types `www.google.com`.

Browser sends query to DNS server.

DNS server replies with Google's IP address.

The browser connects to that IP to load the website.

Command:

`nslookup <domain>` → shows IP address linked to a domain.

5. Proxy Server

Definition:

A **proxy server** acts as an **intermediary** between a user's computer and the internet.

Function:

Routes user requests (like visiting websites) through itself.

Can **filter**, **cache**, or **control** web traffic.

Conceals the user's actual IP address.

Commonly used in schools, offices, and organizations for internet monitoring or security.

Example:

When a user requests to open www.youtube.com, the request first goes to the proxy server → the proxy fetches the page → sends it back to the user.

Advantages:

Enhances privacy and security.

Saves bandwidth via caching.

Controls user access to websites.

Setting Proxy (Windows):

Settings → Network & Internet → Proxy → Manual proxy setup.

Password

Definition:

A **password** is a secret key used to authenticate users and protect access to a computer or network resources.

Importance:

Ensures privacy and data security.

Prevents unauthorized access.

Protects shared files and network resources.

Good Practices:

Use a combination of letters, numbers, and symbols.

Change regularly.

Avoid sharing or writing passwords in visible places.

To Change Password:

Control Panel → User Accounts → Change your Windows Password.

7. User Accounts

Definition:

A **user account** is a record that allows an individual to sign in to a computer and use its resources.

Types:

Administrator Account – full control over system settings and installations.

Standard/User Account – limited access for everyday tasks.

Guest Account – temporary access with minimal privileges.

Importance:

Keeps personal settings and files separate.

Enables multiple people to share one computer securely.

Allows different permissions for users and administrators.

To Manage Users:

Control Panel → User Accounts → Manage another account → Create new account.

Networking:

Each user account can be linked with **network credentials** for shared drives or domain login.

Distributed System

Definition : A **Distributed System** is a collection of **independent computers** that appear to the users as a **single coherent system**.

These computers communicate and coordinate with each other through a **network** to achieve a common goal.

Characteristics

Resource Sharing – Hardware, software, and data are shared among multiple systems.

Concurrency – Multiple processes can execute simultaneously.

Scalability – The system can easily grow by adding more computers.

Fault Tolerance – Even if one computer fails, others can continue functioning.

Transparency – Users view the system as a single unit, even though resources are distributed.

Advantages

Increased **reliability** and **availability**.

Better **performance** through parallel processing.

Cost-effective – Uses inexpensive computers instead of a large mainframe.

Flexibility – Easy to expand and modify.

Disadvantages

Complexity in design and management.

Security concerns due to multiple nodes.

Synchronization and data consistency issues.

Examples

Internet-based applications (e.g., Google Search, YouTube).

Distributed databases. , Cloud computing systems.

Blockchain systems.



Types of Networks

Computer networks are classified based on their **geographical coverage** and **purpose**.

a) LAN (Local Area Network)

Definition: A network that covers a **small area**, such as a room, building, or campus.

Examples: School, college, or office network.

Data transfer speed: Very high (up to 10 Gbps or more).

Ownership: Usually owned and managed by a single organization.

Mediums used: Ethernet cables (wired) or Wi-Fi (wireless).

Advantages:

High data transfer speed.

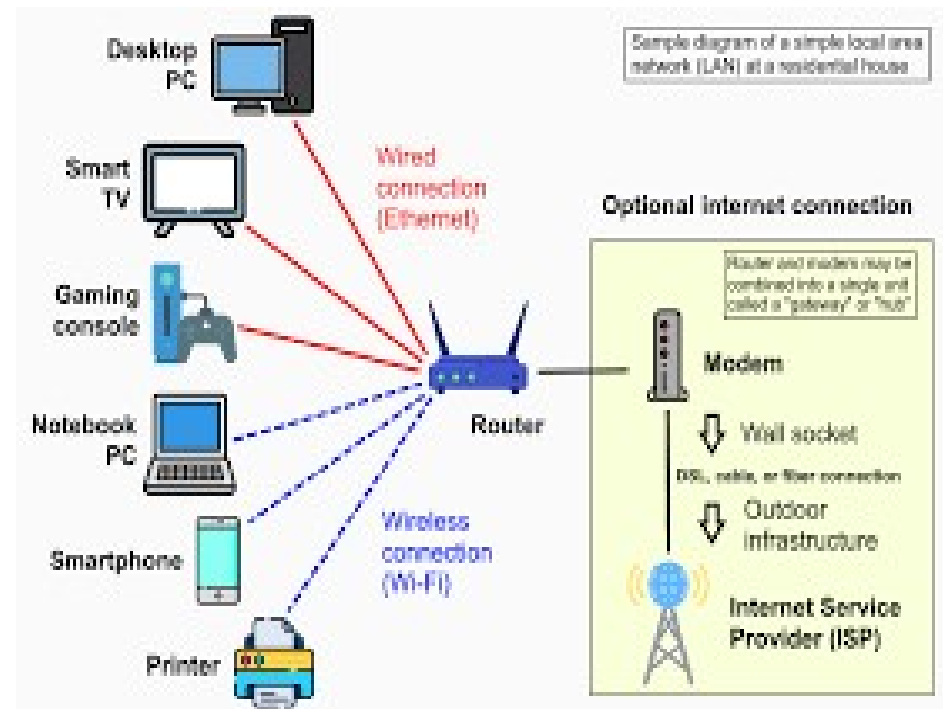
Low error rate.

Easy to set up and maintain.

Disadvantages:

Limited to a small area.

Security issues if not protected.



MAN (Metropolitan Area Network)

Definition: A network that connects multiple LANs in a **city or metropolitan area**.

Examples: Cable TV network, city-wide internet service.

Coverage: 10 to 50 km.

Mediums used: Fiber optic cables, microwave links.

Advantages:

High-speed connectivity between distant LANs.

Useful for organizations spread across multiple city locations.

Disadvantages:

High setup cost.

Maintenance requires skilled professionals.



WAN (Wide Area Network)

Definition: A network that spans **large geographical areas**, even **countries or continents**.

Examples: Internet, international banking network.

Mediums used: Satellite links, optical fiber, leased telephone lines.

Advantages:

Enables global communication.

Supports large organizations.

Disadvantages:

Expensive to install and maintain.

Data transmission may be slower than LAN or MAN.

Prone to security threats.

Components Used to Form a Network

a) Network Interface Card (NIC)

A hardware component that connects a computer to a network.

Can be **wired (Ethernet)** or **wireless (Wi-Fi)**.

b) Switch

Connects multiple computers within a LAN.

Forwards data only to the destination device, improving efficiency.

c) Hub

A simple device that broadcasts data to all connected devices.

Used in small, simple networks.

d) Router

Connects different networks (e.g., LAN to the Internet).

Determines the **best path** for data transmission.

e) Modem

Converts **digital signals** to **analog signals** (and vice versa) for transmission over telephone lines.

f) Repeater

Amplifies weak signals to extend the range of a network.

g) Gateway

Acts as a **translator** between networks using different protocols.

h) Access Point (AP)

Used in wireless networks to connect wireless devices to a wired LAN.

Network Interface Card (NIC)

Definition

A **Network Interface Card (NIC)** is a **hardware component** that connects a **computer or device to a network**.

It allows the computer to communicate with other devices via **wired (Ethernet)** or **wireless (Wi-Fi)** connections.

Functions

Converts **data from the computer** into signals that can be transmitted over the network.

Assigns a unique **MAC (Media Access Control) address** to identify the device.

Handles **data transmission and reception** between the computer and the network.

Types

Wired NIC: Uses Ethernet cable (**RJ-45 connector**).

Wireless NIC: Uses radio waves to connect (**Wi-Fi**).

USB NIC: External card connected via **USB** port.

Advantages

Provides fast and reliable network connection.

Supports both LAN and Internet access.

Essential for both wired and wireless networks



Hub

Definition

A **Hub** is a **basic networking device** that connects **multiple computers** in a **Local Area Network (LAN)** and allows data to be transmitted between them.

Functions

Works at the **Physical Layer (Layer 1)** of the OSI model.

Broadcasts incoming data to **all devices** connected to its ports, regardless of the destination.

Does **not filter** or manage traffic.

Types

Active Hub: Amplifies (boosts) signals before sending them.

Passive Hub: Simply passes the signal without amplification.

Intelligent Hub: Includes basic management features.

Advantages

Easy to install and inexpensive.

Suitable for small networks.

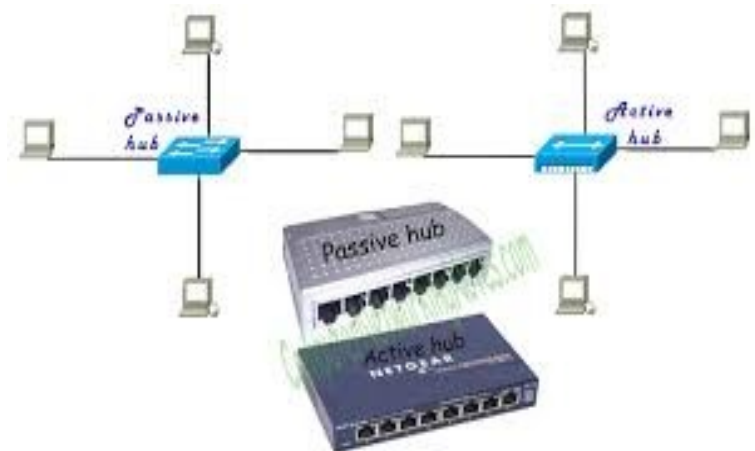
Disadvantages

Creates unnecessary data traffic (broadcasts to all).

Less secure and slower compared to switches.

Example

Used in small LAN setups for connecting 4–8 computers



Switch

Definition

A **Switch** is a **network device** that connects multiple computers and **forwards data only to the intended destination device**, unlike a hub.

Functions

Works at the **Data Link Layer (Layer 2)** of the OSI model.

Uses the **MAC address** of devices to send data efficiently.

Reduces network congestion by avoiding unnecessary broadcasts.

Can manage multiple data transfers simultaneously.

Advantages

Faster than hubs.

Reduces data collisions.

Provides better performance and security.

Disadvantages

More expensive than hubs.

Requires configuration in large networks.

Example

Used in offices or colleges to connect multiple computers within a LAN.



Router

Definition

A **Router** is a **network device** that connects **different networks** (e.g., a LAN to the Internet). It determines the **best path** for data packets to reach their destination.

Functions

Works at the **Network Layer (Layer 3)** of the OSI model.

Routes data packets between **different networks** using **IP addresses**.

Provides **network security** through **firewalls and NAT (Network Address Translation)**.

Supports **Wi-Fi** for wireless devices in home and office networks.

Types

Wired Router: Uses Ethernet cables.

Wireless Router: Provides Wi-Fi connectivity.

Core Router: Used by Internet Service Providers (ISPs).

Edge Router: Connects local networks to external networks.

Advantages

Enables Internet access.

Manages traffic efficiently between networks.

Provides firewall protection.

Disadvantages

More costly than switches or hubs.

Requires configuration knowledge.

Example

Wi-Fi routers used in homes to connect mobile phones, laptops, and TVs to the Internet.



Modem

Definition

A **Modem (Modulator–Demodulator)** is a device that **converts digital signals** from a computer into **analog signals** for transmission over telephone lines — and vice versa.

Functions

Modulation: Converts digital signals (from computer) to analog (for transmission).

Demodulation: Converts received analog signals back into digital form.

Enables Internet connectivity through telephone or cable lines.

Types

Dial-up Modem – Old, slow type using telephone lines.

DSL Modem – High-speed Internet via telephone line.

Cable Modem – Uses TV cables for Internet.

Fiber Modem (ONT) – Used in optical fiber connections.

Advantages

Enables Internet connectivity over existing lines.

Reliable and widely used.

Disadvantages

Dial-up is slow.

Requires ISP connection.



Repeater

Definition

A **Repeater** is a **network device** that **amplifies or regenerates weak signals** to extend the distance over which data can travel.

Functions

Works at the **Physical Layer (Layer 1)**.

Strengthens weak signals and retransmits them.

Used in both **wired and wireless networks**.

Advantages

Increases network range.

Prevents signal loss and distortion.

Essential for large LANs or long cable runs.

Disadvantages

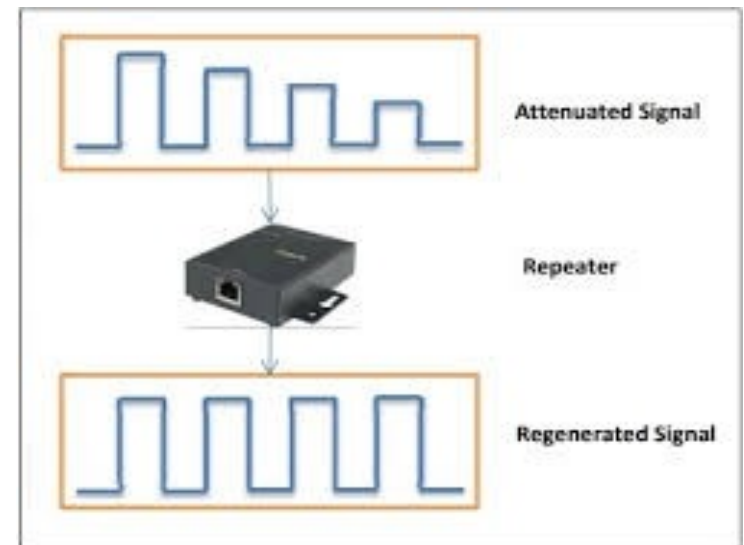
Does not filter traffic.

Cannot connect different networks.

Adds slight delay (latency).

Example

Used in long Ethernet cables or Wi-Fi range extenders.



Summary Table of Networking Devices

NIC – Connects a computer to the network.

Hub – Broadcasts data to all devices.

Switch – Sends data only to the intended device.

Router – Connects multiple networks and routes data.

Modem – Converts digital/analog signals for Internet access.

Repeater – Boosts weak signals to cover long distances.

Device	OSI Layer	Function	Key Feature	Example Use
NIC	Layer 1 & 2	Connects computer to network	Provides MAC address	PC or Laptop network card
Hub	Layer 1	Connects multiple computers	Broadcasts data to all	Small LAN setup
Switch	Layer 2	Forwards data to correct device	Uses MAC address	Office LAN
Router	Layer 3	Connects different networks	Uses IP address	Home/Office Internet router
Modem	Layer 1	Converts digital ↔ analog signals	Enables Internet	Broadband connection
Repeater	Layer 1	Amplifies weak signals	Extends network range	Long cable or Wi-Fi extender

Concepts of Layering

Layering divides the process of data communication into **separate levels (layers)**, where each layer performs a specific function and interacts only with the layers directly above and below it.

Purpose of Layering

Simplifies network design.

Allows interoperability between different hardware and software.

Makes it easier to troubleshoot and update systems.

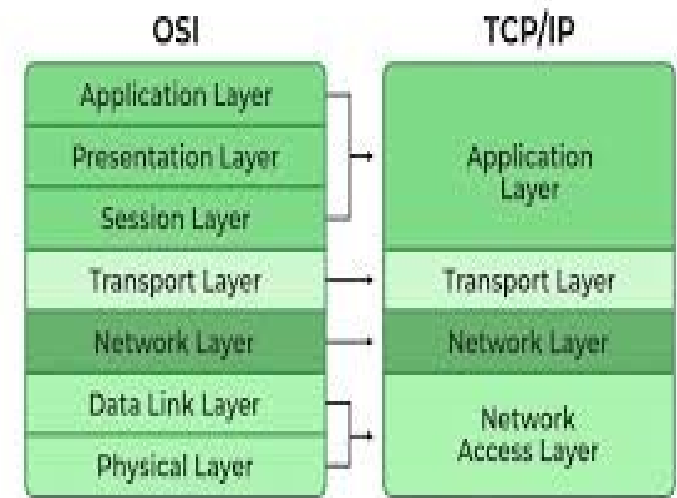
OSI Model (Open Systems Interconnection)

Layer	Name	Function
7	Application	Provides network services to end users (e.g., HTTP, FTP, SMTP).
6	Presentation	Converts data formats, encryption, compression.
5	Session	Manages connections between computers (establish, maintain, terminate).
4	Transport	Ensures reliable data transfer (TCP/UDP).
3	Network	Determines routing (IP addresses).
2	Data Link	Transfers data between adjacent nodes (frames).
1	Physical	Transmits raw bits through cables or wireless media.

TCP/IP Model	Layer	Corresponding OSI Layers	Function
	4	Application (7, 6, 5)	Handles protocols like HTTP, FTP, SMTP.
	3	Transport (4)	Responsible for data transmission (TCP/UDP).
	2	Internet (3)	Handles logical addressing (IP).
	1	Network Access (2, 1)	Deals with hardware and physical transmission.

Comparison of OSI and TCP/IP

Feature	OSI Model	TCP/IP Model
Layers	7	4
Standard	Theoretical Model	Practical Implementation
Protocol Dependency	Protocol Independent	Protocol Specific (TCP, IP)
Usage	Conceptual Learning	Real-world Internet Communication



In Summary

Distributed Systems enable cooperation between multiple independent computers.

Networks (LAN, MAN, WAN) are classified by area coverage.

Topologies define how devices are connected.

Network Components include NICs, routers, switches, and modems.

Layering (OSI, TCP/IP) structures communication logically for reliable data exchange.

What is Network Topology?

Definition:

A **Network Topology** refers to the **physical or logical arrangement** of computers, cables, and other devices in a computer network.

It shows **how devices are connected** and how **data flows** within the network.

2. Types of Network Topologies

There are **two main types** of topologies:

Physical Topology:

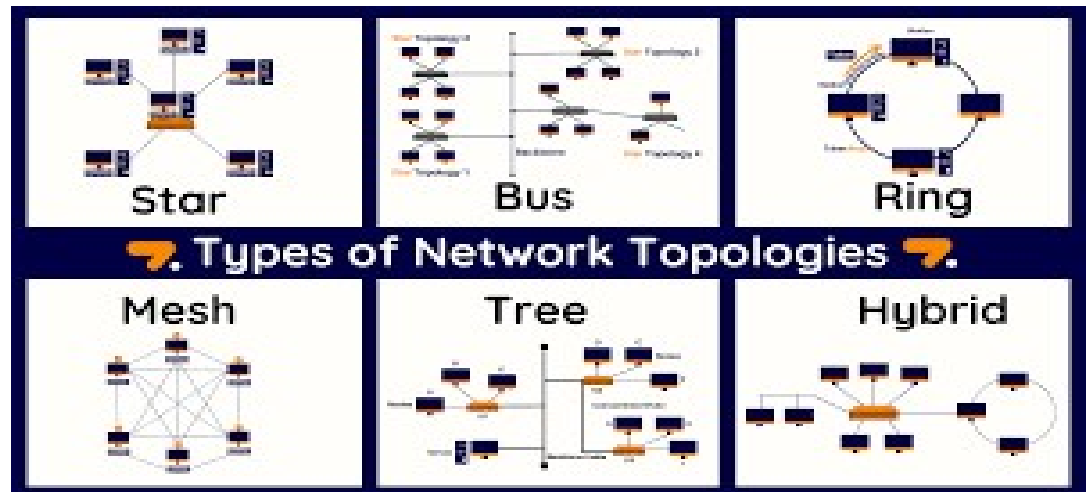
Describes the **actual physical layout** of devices and cables.

Example: Star, Bus, Ring.

Logical Topology:

Describes the **way data travels** within the network, independent of the physical layout.

Example: Ethernet (logical bus), Token Ring (logical ring).



Bus Topology

Description

All devices are connected to a **single central cable**, known as the **bus** or **backbone**.

Data sent from one computer travels in both directions to all other devices.

Advantages

Easy to install and cost-effective for small networks.

Requires less cable length compared to star topology.

Disadvantages

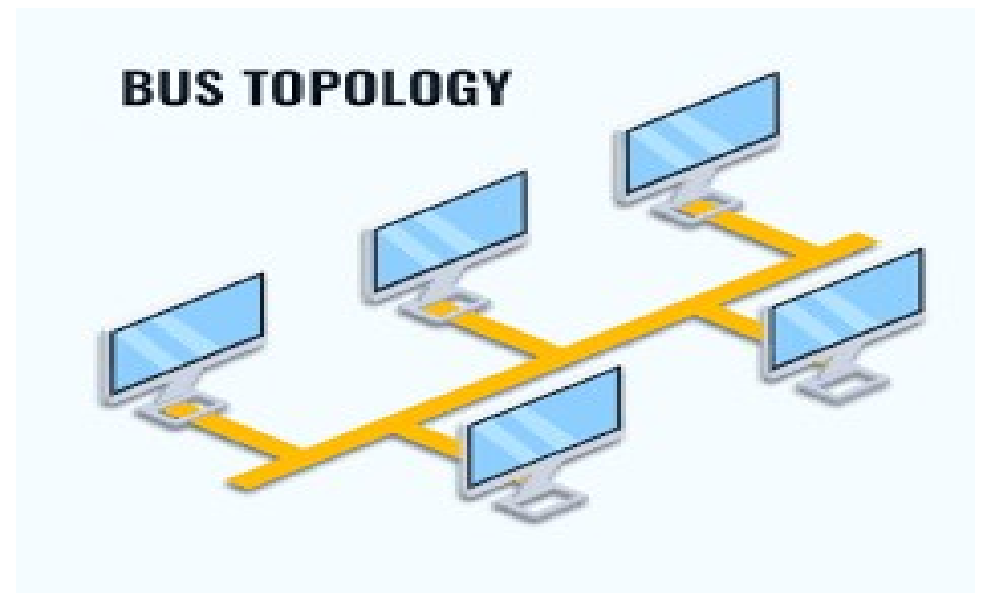
If the main cable (bus) fails, the entire network goes down.

Performance decreases as more devices are added.

Data collisions occur frequently.

Uses

Early LANs and small office networks.



Star Topology

Description

All devices are connected to a **central device** such as a **hub** or **switch**.
Communication occurs through the central device.

Advantages

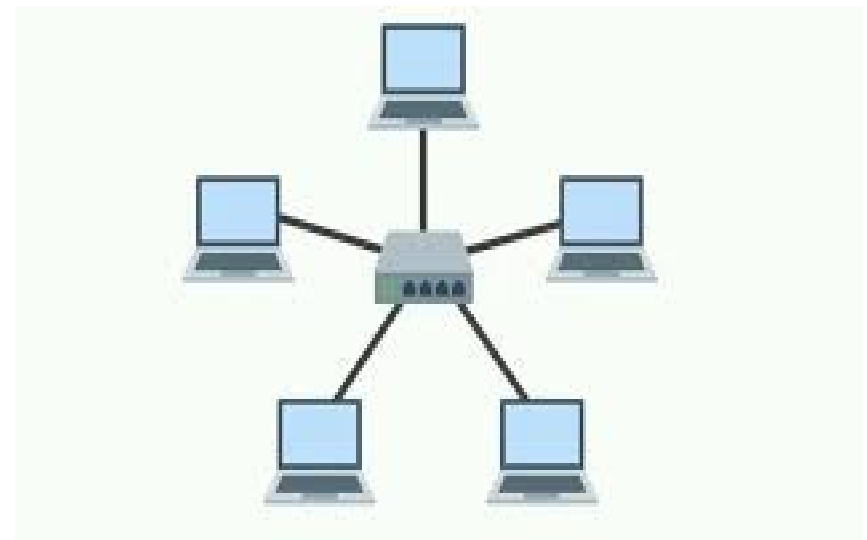
Easy to install and maintain.
Failure of one device does not affect others.
Easy to add new devices.

Disadvantages

If the central hub/switch fails, the entire network stops working.
Requires more cable than bus topology.

Uses

Commonly used in schools, offices, and homes.



Ring Topology

Description

Each device is connected to **two neighboring devices**, forming a **closed loop**.
Data travels in **one direction** (unidirectional) or **both directions** (bidirectional).

Advantages

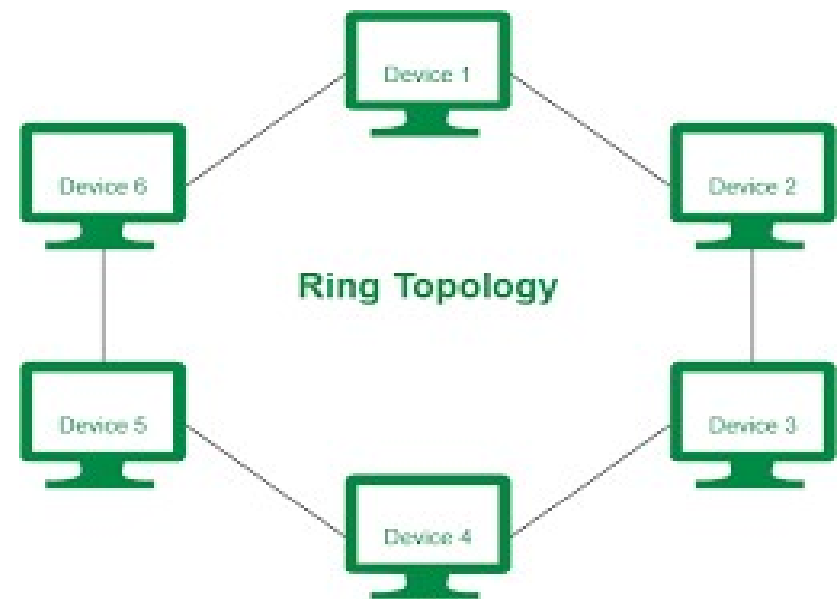
Data transmission is orderly and collision-free.
Performance is consistent even with heavy traffic.

Disadvantages

Failure of one device or cable breaks the loop and stops the entire network.
Difficult to add or remove devices.

Uses

Token Ring networks, fiber optic systems.



Tree Topology

Description

A combination of **Star** and **Bus** topologies.

Multiple star networks are connected using a **main backbone cable**.

Advantages

Scalable — easy to expand the network.

Hierarchical structure simplifies network management.

Fault in one branch doesn't affect others.

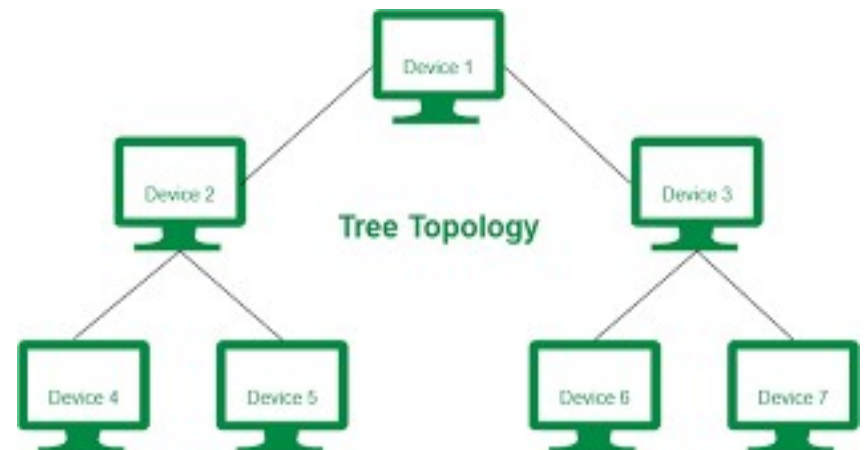
Disadvantages

If the backbone line fails, the entire network is affected.

Installation and maintenance are complex.

Uses

Large organizations, universities, and corporate networks.



Mesh Topology

Description

Every device is connected to **every other device** in the network.

Provides **multiple paths** for data transmission.

Types

Full Mesh: Every device connects to all others.

Partial Mesh: Only some devices are connected directly.

Advantages

Highly reliable and fault-tolerant.

Data can be transmitted via multiple paths.

Network failure does not stop communication.

Disadvantages

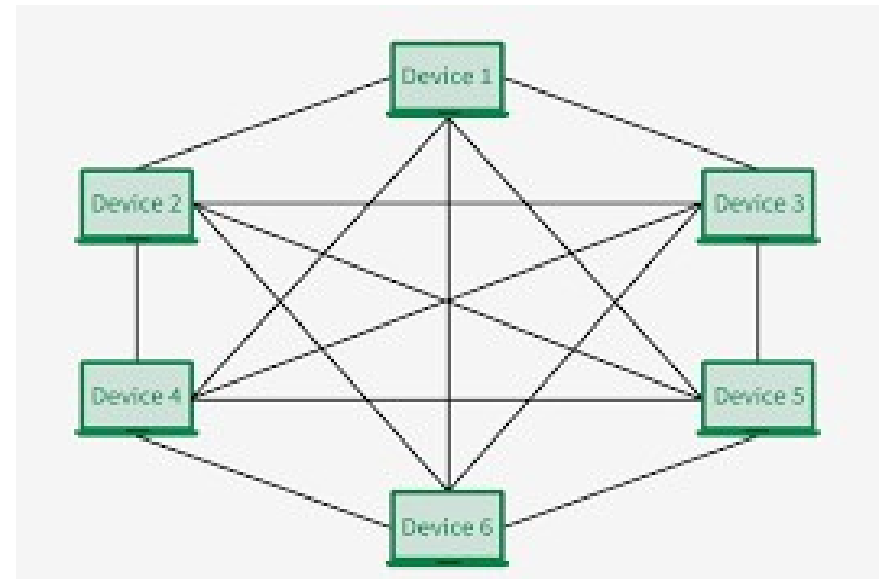
Very expensive due to large cabling.

Complex installation and maintenance.

Uses

Military communication systems.

Data centers and critical networks.



Hybrid Topology

Description

A combination of **two or more** different topologies (e.g., Star + Ring or Star + Mesh).

Advantages

Flexible and scalable.

Designed to meet specific network requirements.

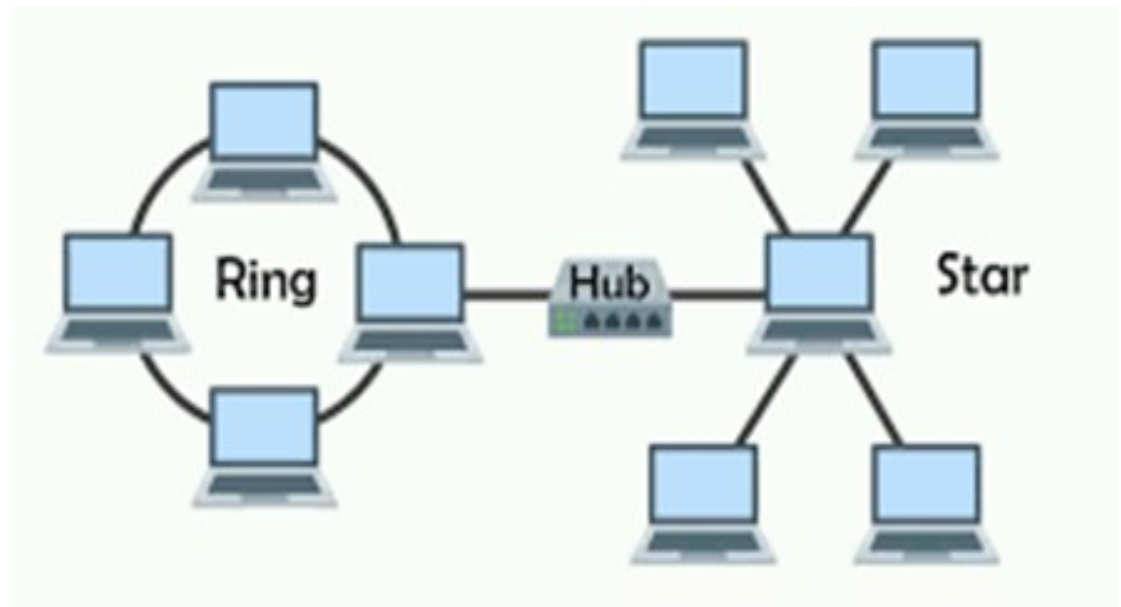
Disadvantages

Expensive and complex to design.

Difficult to manage as the network grows.

Uses

Large enterprises and complex institutional networks.



- **Comparison Table of Topologies**

Topology	Central Device	Cable Required	Reliability	Cost	Ease of Installation	Scalability
Bus	No	Low	Low	Low	Easy	Limited
Star	Yes (Hub/Switch)	Medium	Medium	Medium	Easy	High
Ring	No	Medium	Medium	Medium	Difficult	Low
Tree	Yes	High	High	High	Difficult	High
Mesh	No	Very High	Very High	Very High	Difficult	High
Hybrid	Optional	High	Very High	High	Complex	Very High

Summary

- **Bus Topology** → Simple but prone to failure.
- **Star Topology** → Most popular; reliable and easy to manage.
- **Ring Topology** → Data travels in one direction; fault affects all.
- **Tree Topology** → Hierarchical; suitable for large networks.
- **Mesh Topology** → Highly reliable; very expensive.
- **Hybrid Topology** → Combines strengths of multiple topologies.

Choosing the Right Topology

Key Points to Remember

- Topology affects **network performance, reliability, and cost**.
- **Star topology** is most commonly used today.
- **Mesh topology** is used where **reliability** is critical.
- **Tree and Hybrid** topologies are used in large and complex networks

Requirement

Small office or home network

Simple, low-cost setup

High reliability, no single point of failure

Large and hierarchical organization

Complex enterprise or campus

Recommended Topology

Star

Bus

Mesh

Tree

Hybrid

What is Troubleshooting?

Definition

Troubleshooting is the **process of identifying, diagnosing, and solving problems** in a computer system to make it work properly.

It involves checking **hardware, software, and network components** to detect and fix the cause of malfunction.

Need for Troubleshooting

- To maintain **smooth computer performance**.
- To avoid **system downtime and data loss**.
- To ensure **software and hardware compatibility**.
- To **extend the lifespan** of computer components.
- To improve **system efficiency and reliability**.

Types of Troubleshooting

Type	Description	Examples
Hardware Troubleshooting	Fixing physical components of the computer.	Keyboard not working, no display, loose cable.
Software Troubleshooting	Fixing problems with programs or operating systems.	OS crash, application error, virus infection.
Network Troubleshooting	Resolving connection and communication issues.	Internet not working, IP conflict, cable unplugged.
Peripheral Troubleshooting	Fixing devices connected to computer.	Printer not printing, mouse not responding

Common Computer Problems and Their Causes

Computer Will Not Start

•Possible Causes:

- Loose power cable.
- Faulty power supply (SMPS).
- Damaged motherboard or RAM.

•Solution:

- Check power connection and plug point.
- Test power supply.
- Reseat RAM.

No Display on Monitor

•Possible Causes:

- Loose VGA/HDMI cable.
- Graphics card issue.
- Monitor not powered on.

•Solution:

- Check cable connection.
- Test with another monitor.
- Replace faulty graphics card.

System Freezes or Crashes

Possible Causes:

- Overheating of CPU.
- Faulty RAM.
- Corrupted OS or driver.

Solution:

- Clean CPU fan and heat sink.
- Test RAM using diagnostic tools.
- Reinstall drivers or OS.

Slow Computer Performance

Possible Causes:

- Too many startup programs.
- Virus or malware infection.
- Fragmented hard disk.

Solution:

- Uninstall unnecessary software.
- Run antivirus scan.
- Use disk cleanup and defragmentation.

No Internet or Network Connection

Possible Causes:

- Network cable unplugged.
- Incorrect IP settings.
- Modem or router malfunction.

Solution:

- Check LAN/Wi-Fi connection.
- Reset router.
- Use “Network Troubleshooter” tool.

Printer Not Working

Possible Causes:

- Cable not connected or paper jam.
- Printer driver not installed.

Solution:

- Check printer connection and paper.
- Reinstall printer driver

Steps for Troubleshooting a Computer System

Step	Description
1. Identify the Problem	Observe symptoms or error messages.
2. Analyze the Problem	Check when and how it occurs (startup, while working, etc.).
3. Check Hardware Connections	Ensure cables, devices, and power supply are connected properly.
4. Use Diagnostic Tools	Use inbuilt or third-party tools (Task Manager, Device Manager, etc.).
5. Try Basic Fixes	Restart the computer, update drivers, or perform a system restore.
6. Replace Faulty Components	Replace or repair defective hardware if needed.
7. Test the System	Verify the problem is solved and system is functioning properly.

Tools Used for Troubleshooting

a) Hardware Tools

Screwdriver kit – to open computer cases.

Multi meter – to check power supply voltage.

POST (Power-On Self-Test) Card – to detect motherboard issues.

Spare cables and connectors – to test faulty wires.

b) Software Tools

Device Manager – to check drivers and hardware status.

Task Manager – to monitor CPU and memory usage.

Disk Cleanup & Defragmenter – for performance optimization.

Antivirus / Anti-malware software – to remove infections.

System Restore / Recovery tools – to restore previous system state.

Preventive Maintenance

Preventive maintenance helps **avoid problems before they occur.**

Hardware Maintenance

Clean internal components regularly (dust removal).

Ensure proper ventilation and cooling.

Check power supply and cable connections.

Software Maintenance

Update the operating system and drivers regularly.

Keep antivirus up to date.

Delete temporary and unnecessary files.

Take regular backups.

Sharing of Resources

Definition

Resource Sharing means **allowing multiple computers or users to access shared resources** (like files, printers, and Internet connections) over a network.

Resources That Can Be Shared

Hardware Resources

Printers, scanners, storage devices.

Software Resources

Applications, licensed software.

Data/Files

Shared folders or databases.

Internet Connection

Shared via routers or network switches.

Methods of Sharing Resources

a) In a LAN (Local Area Network)

Files and printers can be shared using:

File and Printer Sharing option in Windows.

Shared folders with permissions.

Network drives mapped to each user.

b) Over the Internet

Cloud storage (Google Drive, OneDrive).

Remote Desktop connections.

Shared servers or **FTP (File Transfer Protocol)**.

c) Using Network Devices

Switch → Connects devices in a LAN.

Router → Connects networks and provides Internet sharing.

Access Point → Enables wireless sharing.

Advantages of Resource Sharing

Reduces cost (one printer for many users).

Increases productivity and collaboration.

Centralized data access and backup.

Easier communication and file transfer.

Disadvantages

Requires network setup and maintenance.

May create **security risks** (unauthorized access).

Performance may reduce if network is overloaded.

Summary

- **Troubleshooting** is a step-by-step process to **detect and solve computer issues**.
- It involves checking **hardware, software, and network components**.

Tools like Device Manager, Task Manager, and multimeters assist in fault detection.

Preventive maintenance minimizes breakdowns and enhances performance.

Resource sharing allows multiple users to share hardware, software, and data efficiently over a network

Difference Between Troubleshooting and Maintenance

Aspect	Troubleshooting	Maintenance
Purpose	Fix existing problems	Prevent future problems
Time of Action	After failure	Before failure
Approach	Reactive	Proactive
Examples	Replacing faulty RAM	Cleaning computer fans regularly